Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

Lecture 5E – Cognition and Human Information Processing II

EIN 3241: Human Factors and Ergonomics
Spring 2026

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Admin Issues

- Midterm on Monday

Divided Attention and Time Sharing

- Process: Multiple Resources

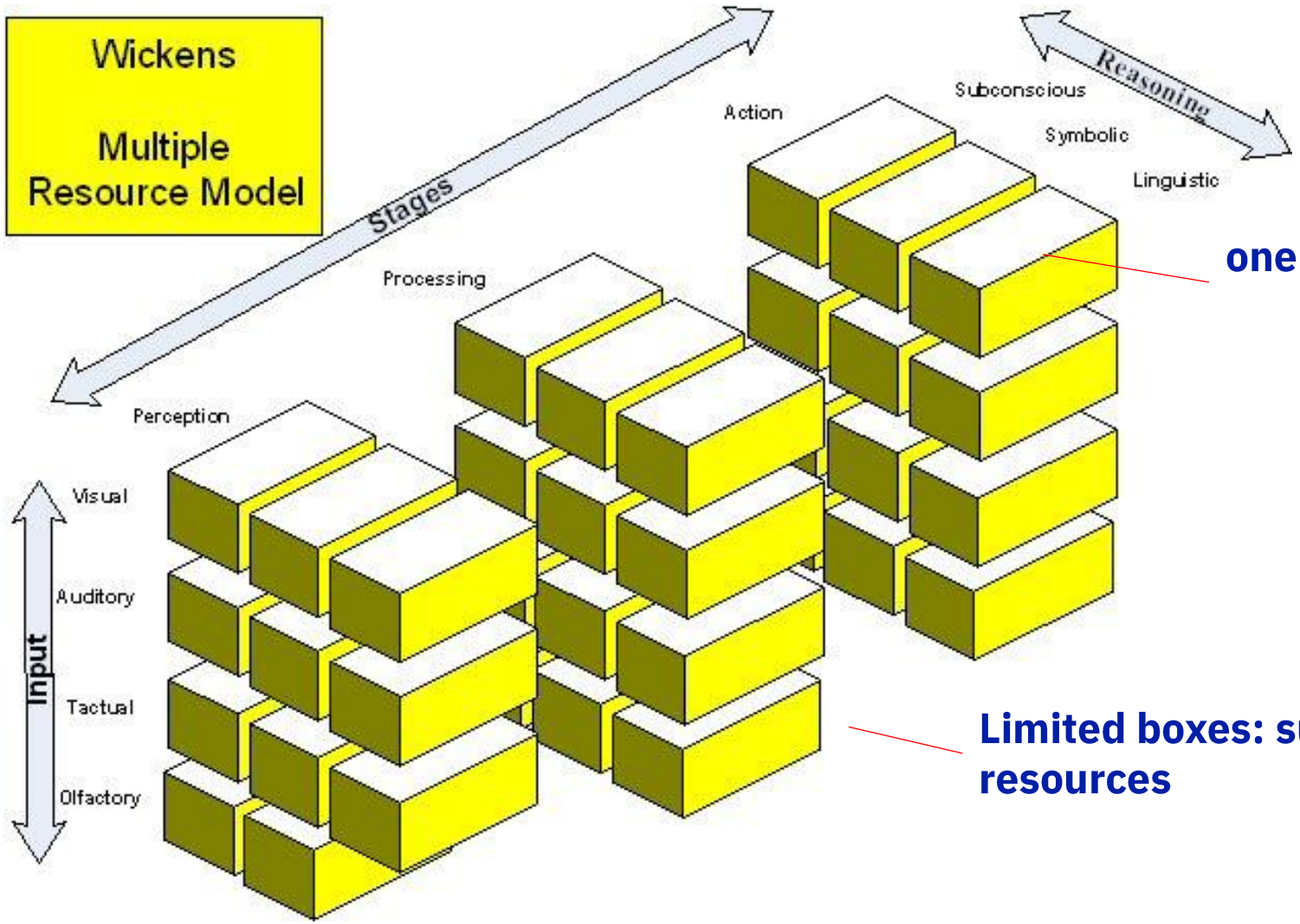
(Wicken's Multiple Resource Theory)

  - Input modality (auditory, visual, tactile, olfactory)
  - Code (spatial, symbolic, linguistic)
  - Stages (perceptual, cognitive, response)

- Each resource can be tapped for an information processing task
- Conflicts when same resource is demanded in parallel

Wickens, C. D. (2008). Multiple resources and mental workload. *Human Factors*, *50*(3), 449-455.

Multiple attentional resources



one cognitive resource

Limited boxes: supply and demand of resources

Attention

- Cognitive capacity = resources
  - Auditory
  - Visual
  - Spatial
- Selective attention
- Focused attention
- Divide our attention to complete several tasks at once
- Attend to both or switch rapidly between two or more tasks
  - While driving?
  - When does the ability to time-share deteriorate?

Stimuli response

- Ergonomists use reaction speed to measure the ability of a person to perform psychomotor tasks.
- **Reaction time (cognitive response):** time that elapses from the appearance of a stimulus to the initiation of movement.
- **Motion time (physical response):** time that elapses during the actual movement.
- **Response time (psychomotor response):** reaction time + motion time

Mental Workload

- Mental workload is one of the most studied constructs from human factors engineering
- Mental workload: perception of effort or demands experienced in the performance of a cognitive task.
- Stress: condition or feeling that demands exceed resources to cope with those demands.
- Stress takes up working memory capacity thereby leaving fewer resources for processing, making it part of the total mental workload.

Mental Workload Measurement

- Mental workload can be determined using three different approaches or their combination:

1. Measures of primary and secondary task performance
2. Physiological measures
3. Subjective workload assessment

1. Measures of primary and secondary task performance

- Primary task performance used as a measure to assess mental workload.
  - Measures may include error rate, speed, reaction times, performance score etc.
- Secondary task performance is assumed to access the spare mental capacity that remains while focusing on the primary task. Poorer performance on secondary tasks implies a more mentally demanding primary task.
  - Measures may include visual detection task, mental arithmetic, recalling information, etc.

2. Physiological measures

- Physiological measures include:
  - Cardiac functions
  - Eye movement (Eye tracking)
  - Electromyography (EMG), i.e. muscle tension
  - Electroencephalogram (EEG), i.e. brain activity
  - Respiration
  - Blood pressure
  - Critical flicker fusion (CFF); 35 40 Hz at resting state; CFF drops 0.5 7 Hz under cognitive load.
- Generally, these measures are more difficult to collect and interpret because the results may not be exclusively affected by mental workload.
- These measures can supplement other workload assessment methods.

2. Physiological measures

- Air traffic controllers:

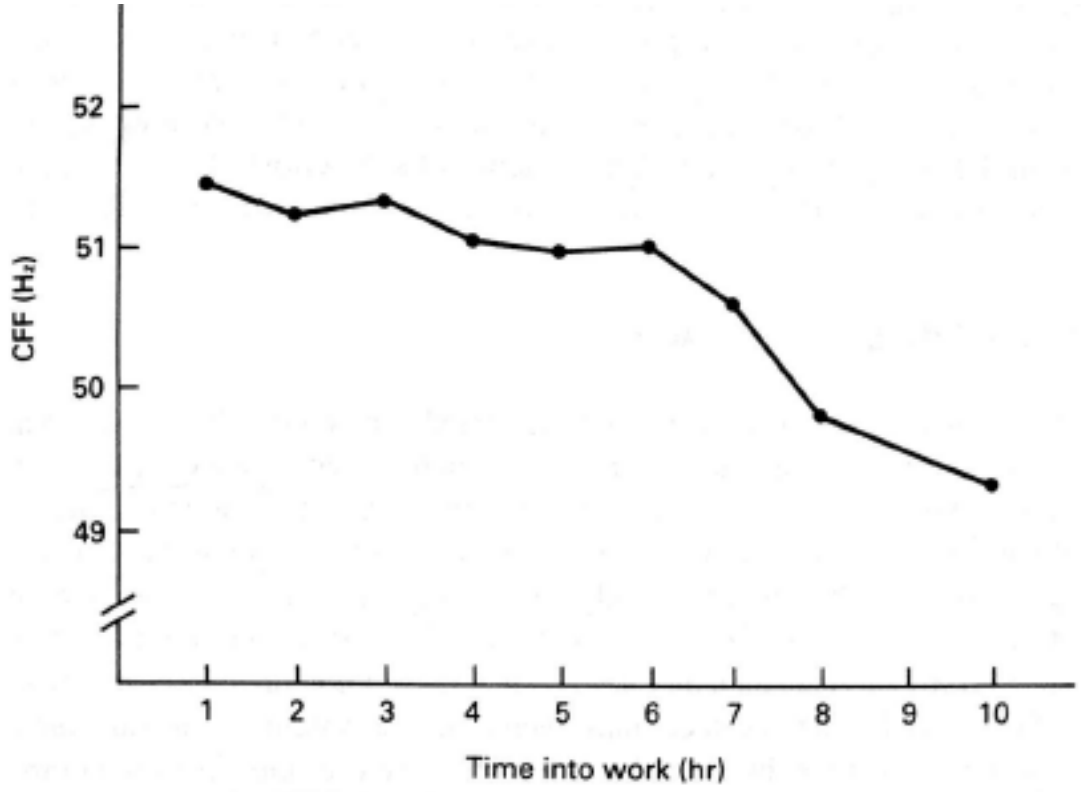


Figure 4.19 CFF drop versus work-hours. (Adapted and reprinted with permission from *Fitting the Task to the Man*, E. Grandjean, 1980. Taylor & Francis, Ltd., London.)

2. Physiological measures

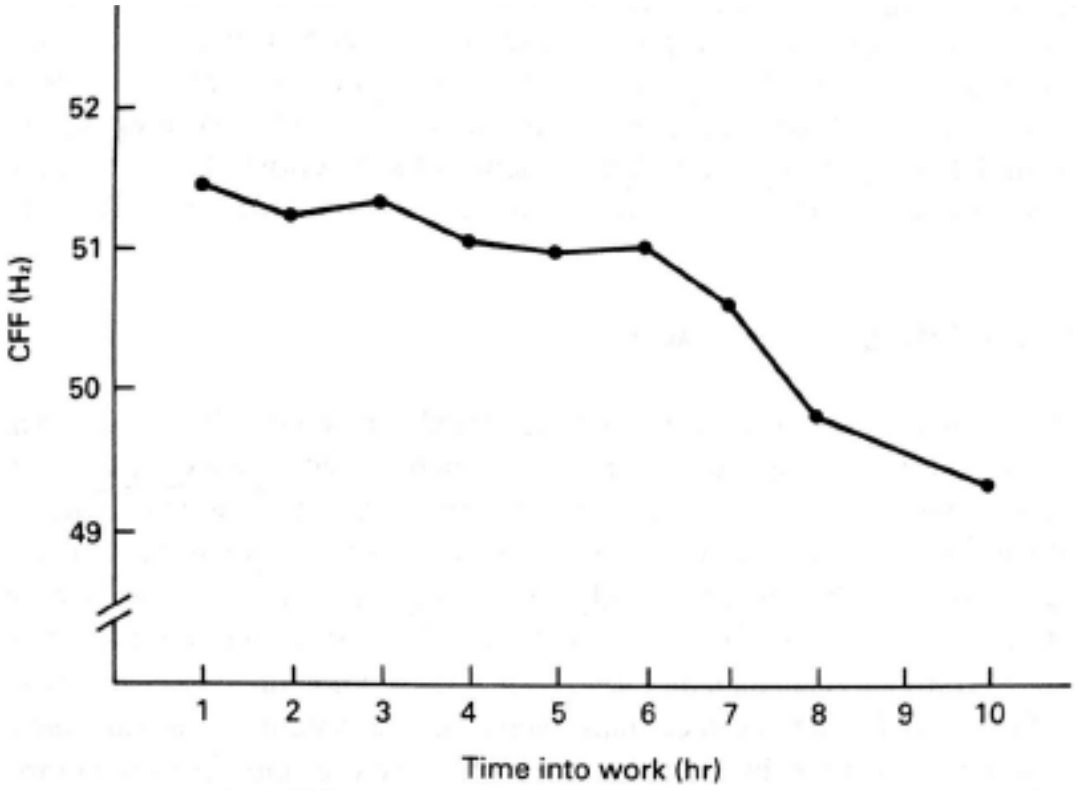


Figure 4.19 CFF drop versus work-hours. (Adapted and reprinted with permission from *Fitting the Task to the Man*, E. Grandjean, 1980. Taylor & Francis, Ltd., London.)

Critical Flicker Fusion (CFF)

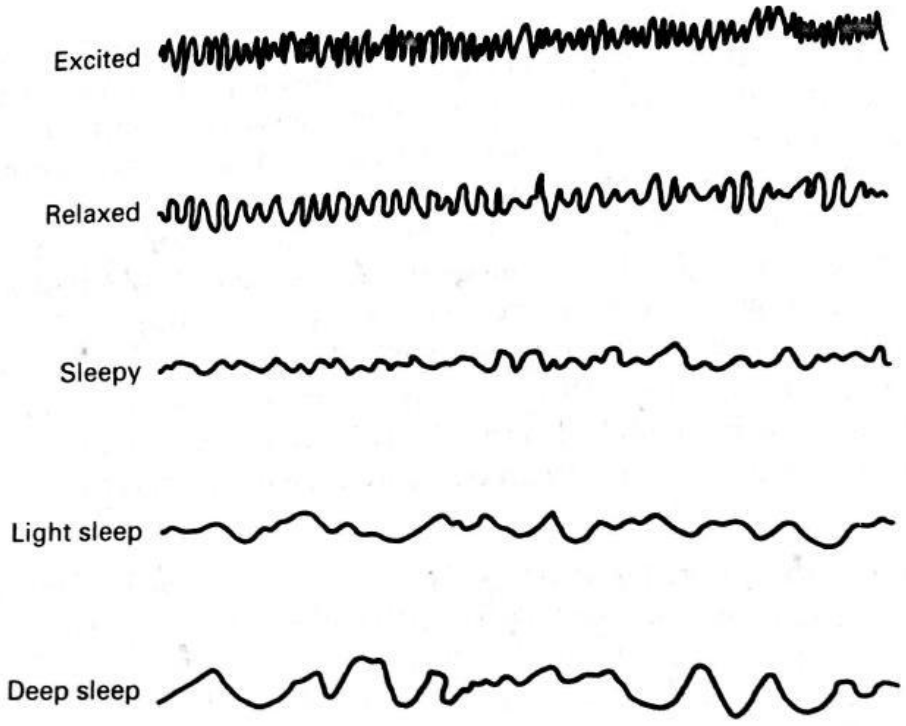


Figure 4.18 EEG tracings for different functional states. (Reprinted with permission from *Fitting the Task to the Man*, E. Grandjean, 1980. Taylor & Francis, Ltd., London.)

EEG

3. Subjective workload assessment

- The subject's direct estimate or comparative judgement of the mental or cognitive workload experienced at a given moment.
- Subjective assessments rely on questionnaires and feedback from employees to determine perceived workload.
- Three commonly used subjective workload assessment measurement approaches in human factors include:
  1. NASA Task Load Index (NASA TLX)
  2. Modified Cooper and Harper Scale (MCH)
  3. Overall Workload Scale (OW)

3. Subjective workload assessment: NASA-TLX

- NASA TLX: a subjective, multidimensional assessment tool for rating perceived workload.
- “Workload” is divided into six dimensions or
  1. Mental demand
  2. Physical demand
  3. Temporal demand
  4. Performance
  5. Effort
  6. Frustration
- Each scale consists of twenty steps ranging from 0 to 100 (ratings are assigned to the nearest 5).

3. Subjective workload assessment: NASA-TLX

TABLE 9.3
NASA-TLX Rating Scale Definitions

Workload Component	Endpoints	Description
Mental demand	Low/high	How much mental and perceptual activity is required (e.g., thinking, deciding, calculating, remembering, looking, searching, etc.)? Was the task easy or demanding, simple or complex, exacting or forgiving?
Physical demand	Low/high	How much physical activity is required (e.g., pulling, pushing, turning, controlling, activating, etc)? Was the task easy or demanding, slow or brisk, slack or strenuous, restful or laborious?
Temporal demand	Low/high	Did you feel the time pressure due to the rate or pace at which the task or task elements occurred? Was the pace slow and leisurely or rapid and frantic or somewhere in between?
Effort	Low/high	How hard did you have to work mentally and physically to accomplish your level of task performance?
Frustration level	Low/high	How insecure, discouraged, irritated, stressed, and annoyed versus secure, gratified, content, relaxed, and complacent did you feel during the task?
Performance	Good to poor	How successful do you think you were in accomplishing the goals of the task set by the experimenter (or yourself)? How satisfied were you with your performance in accomplishing these goals?

Source: Taken directly from Hart, NASA AMES Research Center, 2007.

Name	Task	Date

Mental Demand — How mentally demanding was the task?

Very Low — Very High

Physical Demand — How physically demanding was the task?

Very Low — Very High

Temporal Demand — How hurried or rushed was the pace of the task?

Very Low — Very High

Performance — How successful were you in accomplishing what you were asked to do?

Perfect — Failure

Effort — How hard did you have to work to accomplish your level of performance?

Very Low — Very High

Frustration — How insecure, discouraged, irritated, stressed, and annoyed wereyou?

Very Low — Very High

3. Subjective workload assessment: NASA-TLX

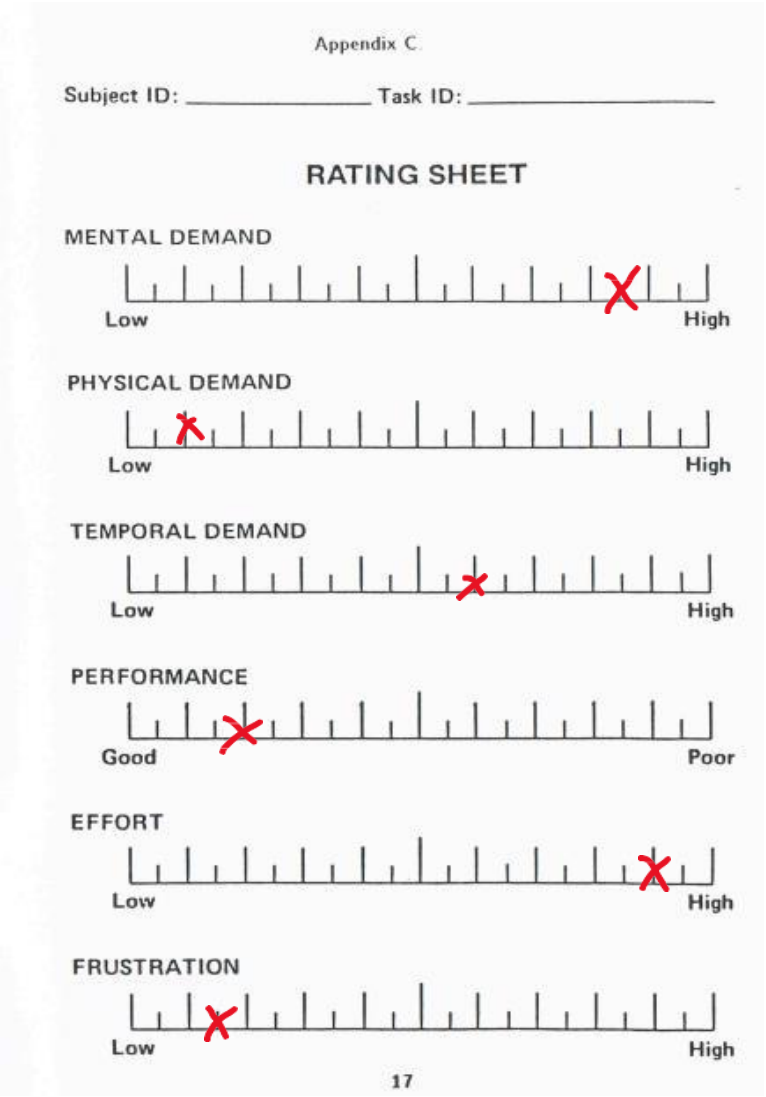
- The task load index (TLX) is an overall workload score derived from the weighted average of the scale ratings.
- TLX score computation:
  1. Determine the individual weighting of each scale by asking participants to compare them pairwise in terms of their perceived importance/relevance to the task (a total of 15 pairwise comparisons).
  2. The total number of times each scale is chosen is the weighted score.
  3. Divide the weighted scores by 15 to get the relative importance of each scale.
  4. Multiply the weight of each scale to its corresponding score and add them together.
- Many researchers use the “Raw TLX” scores, without considering the overall TLX.
  - i.e., individual scales may be dropped if less relevant to the task.

NASA-TLX Example

- Implement the NASA TLX scale to measure the overall workload of a task

Effort or Performance	Temporal Demand or Frustration	Frustration or Effort	Performance or Mental Demand
Temporal Demand or Effort	Physical Demand or Frustration	Performance or Temporal Demand	Mental Demand or Effort
Performance or Frustration	Physical Demand or Temporal Demand	Mental Demand or Physical Demand	Effort or Physical Demand
Physical Demand or Performance	Temporal Demand or Mental Demand	Frustration or Mental Demand	

15 16



Appendix C

Subject ID: _____ Task ID: _____

RATING SHEET

MENTAL DEMAND

Low High

PHYSICAL DEMAND

Low High

TEMPORAL DEMAND

Low High

PERFORMANCE

Good Poor

EFFORT

Low High

FRUSTRATION

Low High

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NASA-TLX Example

Appendix D.

Subject ID: _____ **Date:** _____

SOURCES-OF-WORKLOAD TALLY SHEET

Scale Title	*Tally*	*Weight*
MENTAL DEMAND	IIIII	5
PHYSICAL DEMAND		0
TEMPORAL DEMAND	II	2
PERFORMANCE	III	3
EFFORT	IIII	4
FRUSTRATION	I	1

Total count = 15

(NOTE - The total count is included as a check. If the total count is not equal to 15, then something has been miscounted. Also, no weight can have a value greater than 5.)

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Appendix E.

Subject ID: _____ **Task ID:** _____

WEIGHTED RATING WORKSHEET

Scale Title	*Weight*	*Raw Rating*	*Adjusted Rating* (Weight X Raw)
MENTAL DEMAND	5	85	425
PHYSICAL DEMAND	0	10	0
TEMPORAL DEMAND	2	60	120
PERFORMANCE	3	20	60
EFFORT	4	90	360
FRUSTRATION	1	15	15

Sum of "Adjusted Rating" Column = 980

WEIGHTED RATING = [i.e., (Sum of Adjusted Ratings)/15] 65.3

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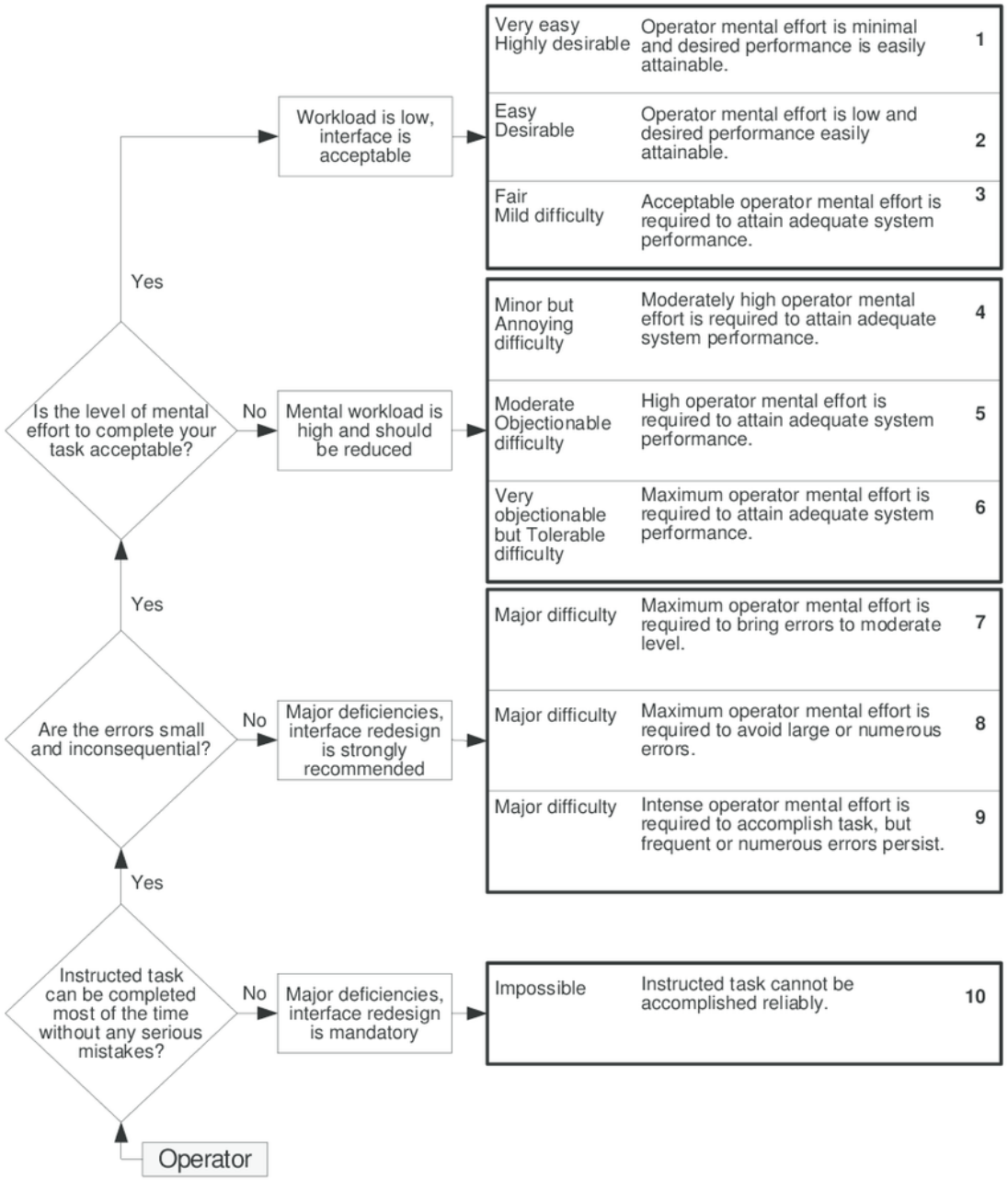
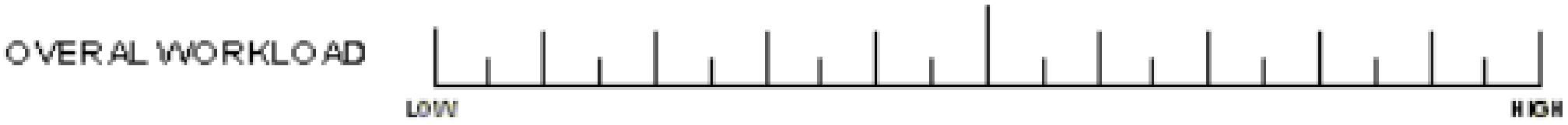
3. Subjective workload assessment: Modified Cooper and Harper Scale (MCH)



3. Subjective workload assessment: Overall Workload scale (OW)

- Overall workload: is a rating of the subject's overall workload on a unidimensional scale of 0 to 100 (assigned to the nearest 5).
  - 0 = very low workload
  - 100 = very high workload
- A single, 20 step bipolar scale is used to obtain this global rating.
- "The total workload associated with the task considering all sources and components."



Comparing the Three Techniques

- MCH and OW are unidimensional workload scales, which take less time to collect and analyze the data.
- TLX is a multidimensional scale, which entail special sorting procedures that take more time, but are justified by the detailed, diagnostic information that they provide.
- TLX and OW are consistently superior in terms of sensitivity and have the strongest operator acceptance.

Other Psychophysical Tools

- Borg scale
- Likert scale
- Questionnaire design
- Visual Analog scale (VAS)
- …

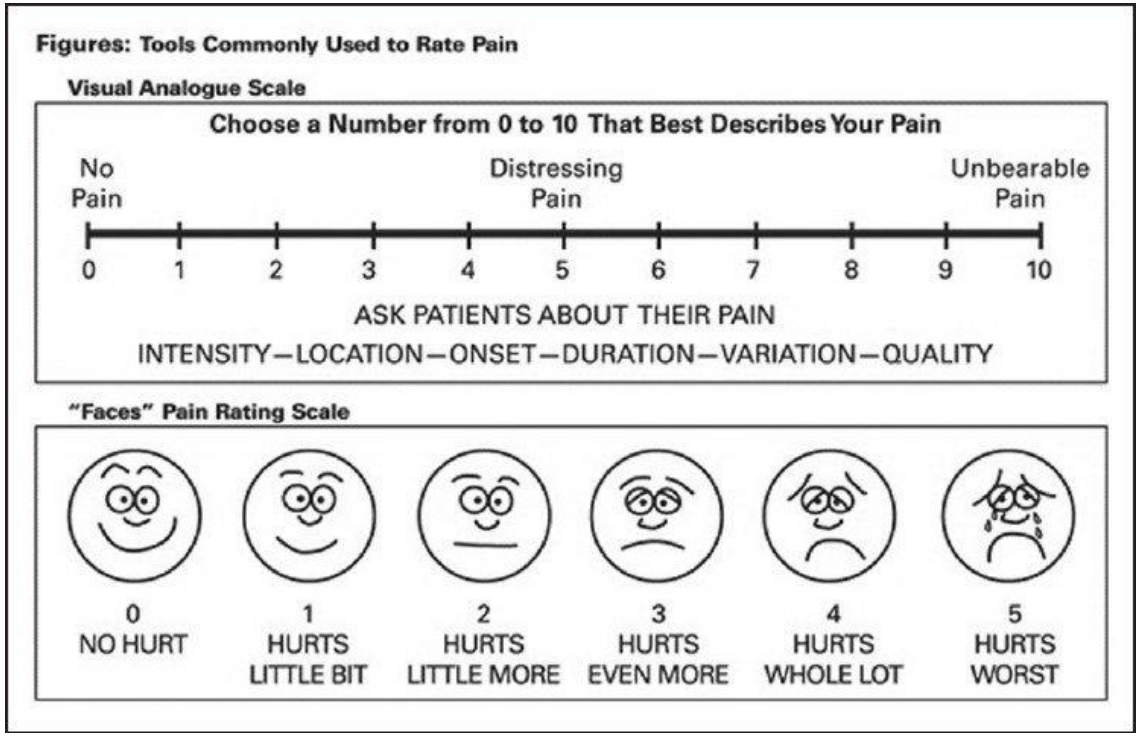
Borg RPE Scale

- The Borg Rating of Perceived Exertion (RPE) is a way to subjectively measure the physical activity intensity level.
- Perceived exertion is how hard you feel like your body is working.
  - Increase in heart rate
  - Increase in breathing rate
  - Increase sweating;
  - Muscle fatigue

BORG RPE Scale	
Scoring	**Level Of Exertion**
6	No Exertion
7	Extremely Light
8	
9	Very Light
10	
11	Light
12	
13	Somewhat Hard
14	
15	Hard Heavy
16	
17	Very Hard
18	
19	Extremely Hard
20	Maximal Exertion

Visual Analogue Scale

- The visual analogue scale (VAS) is a subjective scale to measure the level of pain and to evaluate the mood of the patient.
- Is often used in clinical research.



Likert Scale

- Likert scale is a rating scale used to measure a person's level of agreements.
  - Ease of use of a product
  - Usefulness of a product
  - Ease of maintenance
  - Ease of learning

Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1	2	3	4	5

UF

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LEADING THE CHARGE, CHARGING AHEAD